Chapter 1

Diminutives of English irregular plurals: Morpho-phonological evidence for late merge within the word

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English has a diminutive suffix /-i/ which appears between the noun and plural suffix (*doggies, birdies*). English also has nouns which show umlaut in the plural (*goose/geese, mouse/mice*). I observe that for such nouns, the diminutive forces use of the regular plural suffix /-z/, but permits optional umlaut (*goosies/geesies*). I take umlaut in such nouns to involve synchronic suppletion, which interacts with plural morpho-phonology in different ways, depending on whether the diminutive suffix is merged into the structure of the word earlier, or later in the derivation. This analysis builds on research into the possibility of late merger, as well as hypotheses about adjacency in suppletion, and the bottom-up nature of morpho-phonological determination in syntactic structures. Ultimately, while the patterns examined here might at first glance appear idiosyncratic, this analysis shows that they in fact emerge for principled reasons, which are informative for linguistic theory.

Keywords: English, diminutive, plural, umlaut, late-merge, limits of spell-out

1 Introduction

I analyze certain morpho-phonological patterns in English plural nouns that include the diminutive suffix /-i/. The orthographic form of this morpheme varies, generally being *-(e)y* or *-ie*. This diminutive is highly productive in spoken English, and is common in colloquial and child-directed speech.

(1) *The diminutive /-i/*

a. Look at the cute **doggy**.
b. Some **birdies** live in this tree.
c. There's a little **fishie** in the pond.

This diminutive encodes that the noun in question is small and/or endearing. It is usable with both inanimate and animate nouns, including names. We most often find this diminutive used with mono-syllabic nouns:

(2) *Diminutive of mono-syllabic nouns*

a. horse → horsie
b. sheep → sheepie
c. foot → footie
d. snack → snackie
e. Jim → Jimmy

English has several other diminutive forms that are likely not fully productive, which I set aside here.[1] For a general overview of English diminutives, see Huddleston & Pullum (2002) and Schneider (2003).

In this chapter, I focus on the behavior of diminutive /-i/ with plural nouns. In regular plurals, the diminutive suffix sits between the noun and the plural suffix:[2]

(3) *Diminutive with regular plurals*

a. dogs → dogg-<u>ie</u>-s
b. pigs → pigg-<u>ie</u>-s
c. birds → bird-<u>ie</u>-s

[1]Here are some examples of such diminutives:

(i) a. *-(l)et(te)*: drop**let**, pig**let**, cigar**ette**
b. *-ling*: gos**ling**, seed**ling**, dar**ling**

[2]This diminutive suffix may have an allomorph /-zi/ with nouns that end in vowels, perhaps including glides (p.c. Jane Lorenzen, Andrew Murphy). Interestingly, this version of the diminutive only seems to be possible in plural contexts:

(i) a. toe → two toesies / *one toesie
b. fly → two fliesies / *one fliesie
c. shoe → two shoesies / *one shoesie

This second /z/ resembles an additional exponent of plurality, at least superficially. Understanding the status of such forms requires further work. In the first place, it is unclear how productive such diminutives actually are.

English also has irregular plural nouns. These do not use the regular plural suffix /-z/, and many of these also involve a vowel shift, known as *umlaut*. For an overview see Hejná & Walkden (2022) ch. 6.2.2. For example, the noun *mouse* pluralizes with umlaut to *mice*, which lacks an overt plural suffix. However, I observe that the diminutive plural of this and similar nouns must exceptionally use the regular plural suffix, and if plural umlaut would normally occur with that noun, umlaut is permitted though not required, as (4) previews.

(4) *Preview: Diminutive of irregular plural nouns*

a. *Normal singular*
one mouse

b. *Normal plural*
two mice / *two mouses

c. *Diminutive singular*
one mousie

d. *Diminutive plural*
two **mousies** / two **micies**

The same patterns can be found with nouns such as *goose, tooth, foot*, and so on. The full range of data is discussed in the following section. The goal of this chapter is to explain this pattern, situating it within relevant theoretic research.

The term *umlaut* is often used to refer to a diachronic process. However, here I use it to refer to this vowel alternation as we see it in present day English. I propose that this umlaut is synchronically encoded in English grammar as a form of suppletion, as we will see. While nouns that show umlaut are the focus of this chapter, this analysis will also naturally extend to diminutives of irregular plurals in English more generally.

The patterns reported here have been corroborated by judgments gathered from 12 native speakers, through a mixture of online surveys and in-person discussions (and many more through informal conversations). This includes four British speakers, two Australian ones, the remainder being Americans. While some speakers do not readily accept diminutives of irregular plurals at all (particularly two American speakers of those consulted), it is clear that many do, within the appropriate linguistic register.[3] However, here I abstract away from the fact that speakers vary in their acceptance of particular forms. Specifically, around half of speakers remarked that irregular plural diminutives with umlaut, for certain nouns, are less acceptable than equivalents that lack umlaut. In the

[3]The Australian individuals note that diminutives and analogous truncated forms are very common in Australian English.

context of this chapter's analysis, this could stem from the fact that diminutives with umlaut require a special operation—late merger of an affix, as we will see. However, such judgments are not systematic. This is not surprising, since the phenomenon in question exists on the periphery of English grammar. This fact, along with its marked informal/colloquial nature, makes systematic data collection challenging. These limitations mean that there is an opportunity for more careful empirical investigation. The general pattern is nevertheless clear.

I argue that these facts emerge from the way that the morpho-phonological realization of a diminutive noun interacts with its syntactic derivation. I make this analysis concrete using the Distributed Morphology theory (Halle & Marantz 1993, Harley & Noyer 1999, and many more), which facilitates straightforward statements about the interplay between these domains of grammar. This theory posits that words contain syntactic structure, which is assigned phonemic content at its terminal nodes via a language-specific list of *vocabulary insertion* (VI) rules. Additionally, I adopt two general hypotheses that are common in research using Distributed Morphology. The first is the proposal that the assignment of morpho-phonological information to a structure proceeds bottom-up, starting at the root (Embick 2010, Bobaljik 2000, 2012). The second is the hypothesis that for one syntactic element to trigger use of a context-sensitive VI rule for another, and hence yield context-sensitive suppletion/allomorphy, the two must be adjacent (Embick 2010, Bobaljik 2012, Bobaljik & Harley 2017).

I combine these hypotheses with a proposal about the English diminutive. Previous works have argued that diminutive affixes in some languages are morphologically modifier-like, analogous to adjuncts in syntax. See for instance Fábregas (2013), Gouskova & Bobaljik (2022), and references therein. Furthermore, research in syntax and semantics has argued that adjuncts/modifiers can be introduced into the derivation at a relatively late point (Lebeaux 1991, Sauerland 1998, Nissenbaum 2000, Fox 2002). Late attachment in morphology also has some precedent (Pesetsky 1985, Nissenbaum 2000, Newell 2005, 2008). Combining these concepts, I argue that the English diminutive /-i/ is a morpho-syntactic adjunct, which thus can be late-merged into the word. I show that this proposal, when combined with the hypotheses from Distributed Morphology summarized above, predicts the patterns in English plural diminutives.

Ultimately, I argue that earlier merger of the diminutive yields forms without umlaut, and later merger of the diminutive derives forms with umlaut. This analysis, if correct, unites findings from multiple domains of linguistic theory to provide a principled explanation for what might otherwise be regarded as an isolated idiosyncrasy of colloquial English.

1.1 Contents of the chapter

Next, section 2 discusses the data in detail. Section 3 provides background on late merge, its relevance to diminutive morphemes, and my implementation of it. In section 4, I show that earlier merger of the diminutive suffix creates plural diminutives without umlaut. In section 5, I demonstrate that later merge of the diminutive derives forms with umlaut. The bottom-up nature of the VI process, and an adjacency requirement on context-sensitive suppletion, are essential to the analysis in these sections. Furthermore, section 6 shows that a proposed limitation on late merge from Nissenbaum (2000) accurately constrains the predictions. Section 7 extends this analysis to an account for certain instances of long-distance allomorphy. Section 8 concludes, and describes a remaining puzzle about the forms *kitty* and *bunny*.

2 The facts about diminutives of irregular plurals

Here I describe in more detail the facts about diminutive irregular plurals. As mentioned above, the diminutive suffix /-i/ is most commonly affixed to monosyllabic nouns, yielding a disyllabic result. Indeed, Schneider (2003) states that forms using diminutive /-i/ are always disyllabic, forming a trochaic prosodic foot. See also Kempe et al. (2005). Lappe (2002) makes a similar observation about certain truncated diminutive-like names in English. Thus there appears to be a phonological requirement for such diminutive forms to be disyllabic.[4] It

[4] Many speakers tolerate examples where the diminutive affixes to a multi-syllabic noun that carries final stress, in which case the diminutive is perhaps licit due to ending up as part of a trochee, building on Schneider's generalization:

(i) a. giraffe ([ʤɪ.ˈɹæf]) → giraffie
 b. raccoon ([ɹæ.ˈkun]) → racoonie

While affixation of the diminutive suffix to a noun that does not end in a stressed syllable fails to form a trochee and is unacceptable, in many cases truncation of the noun results in a more acceptable result:

(ii) a. vegetable ([ˈvɛʤ.tə.bl]) → *vegetablie / ✓ veggie
 b. blanket ([ˈbleɪŋ.kɪt]) → *blanketie / ✓ blankie

These effects could be analyzed as the result of an alignment constraint (McCarthy & Prince 1993) that requires the diminutive suffix to be adjacent to a stressed syllable, such that it is at the end of a trochee. This must indeed be a property unique to the diminutive suffix, because the homophonous adjectival suffix /-i/ is not subject to this requirement. This suffix can therefore attach to nouns that do not end in a stressed syllable (iii):

turns out that the irregular plural nouns that are relevant for this investigation are all mono-syllabic, and thus yield disyllabic forms when combined with the diminutive suffix. Thus there is no phonological confound regarding affixation of diminutive /-i/ to them. We can therefore proceed without considering this factor further.

As mentioned above, with nouns that use regular plural morphology, the diminutive sits between the noun and the plural suffix:

(5) *Diminutive with regular plurals*

a. dogs → dogg-ie-s

b. pigs → pigg-ie-s

c. birds → bird-ie-s

The same is true in the Slavic languages, for instance, where diminutives are very frequent (Moskal 2015, Gouskova & Bobaljik 2022). For English nouns that are irregular in the plural, the patterns are more complex. The irregular nouns relevant here do not use the regular plural suffix /-z/, but instead either replace it with an alternative suffix (6a-b) or simply omit it (6c-h). Among the nouns that omit the plural suffix, several of them show umlaut (6c-g):[5]

(6) *Relevant examples of irregular plurals*

a.	one ox / two ox-en (*oxes)	[Alternative plural suffix]
b.	one child / two child-ren (*childs)	[Alternative plural suffix]
c.	one mouse / two mice (*mouses)	[Umlaut, no plural suffix]

(iii) a. Drinking water with your nose is a very **elephanty** thing to do.
b. John hasn't cleaned his hamster's cage, so his room has a **hamstery** smell.

This could be taken as evidence for morpheme-specific phonological constraints (Pater 2000, 2009) that distinguish the homophonous adjectival and diminutive suffixes. Alternatively, under cophonology theory (Orgun 1996, Inkelas & Zoll 2007, Sande et al. 2020), we could posit that these suffixes are associated with separate phonological grammars, which contain distinct phonological constraints, but no explicit link between phonological constraints and morphemes. Either type of analysis is sufficient for the purposes of this chapter.

[5]Two other nouns that display umlaut, which I do not address, are *man/men* and *woman/women*. A diminutive **womenie* is independently ruled out due to violating the requirement for such diminutives to be disyllabic. A diminutive such as **manie* is absent from the author's English and thus was not tested, but this is evidently available in Scottish English (see https://www.merriam-webster.com/dictionary/mannie). In principle, this chapter's analysis predicts a speaker who allows such a form to pluralize it as either *mannies*, without umlaut, or *mennies*, including umlaut. I leave this for future research.

d. one louse / two lice (*louses) [Umlaut, no plural suffix]
e. one goose / two geese (*gooses) [Umlaut, no plural suffix]
f. one tooth / two teeth (*tooths) [Umlaut, no plural suffix]
g. one foot / two feet (*foots) [Umlaut, no plural suffix]
h. one sheep / two sheep (*sheeps) [No umlaut, no plural suffix]

The diminutive suffix has a complex interaction with these nouns. One possibility is for the diminutive to block all irregular plural morphology, as (7) shows. In this case, umlaut does not apply, and the regular plural suffix /-z/ appears. This obligatorily replaces what we might regard as a null plural suffix in nouns like *mice, geese* and so on. The intuition that such nouns normally have a null plural suffix, which is overridden by the default suffix /-z/ in diminutive forms, will be essential to the coming analysis. One noun here has an overt irregular plural suffix (*ox-en*) which is similarly replaced by /-z/ when the diminutive is present. Finally, various nouns involve a null plural suffix and never show umlaut (*sheep, fish*), and like the other irregular nouns, these receive the suffix /-z/ as diminutive plurals. Overall, the generalization is that in all diminutive irregular plural nouns, umlaut (if normally applicable) need not occur, and whatever plural suffix would normally be present (even if null) is replaced by /-z/:

(7) *One possibility: All irregular morphology blocked by diminutive*
a. ox-en → ✓ox-$\underline{\text{ie}}$-$\underline{\text{s}}$ / *ox-ie-en
b. mice-$\varnothing_{\text{PL}}$ → ✓mous-$\underline{\text{ie}}$-$\underline{\text{s}}$ / *mice-ie-$\varnothing_{\text{PL}}$
c. geese-$\varnothing_{\text{PL}}$ → ✓goos-$\underline{\text{ie}}$-$\underline{\text{s}}$ / *geese-ie-$\varnothing_{\text{PL}}$
d. feet-$\varnothing_{\text{PL}}$ → ✓foot-$\underline{\text{ie}}$-$\underline{\text{s}}$ / *feet-ie-$\varnothing_{\text{PL}}$
e. teeth-$\varnothing_{\text{PL}}$ → ✓tooth-$\underline{\text{ie}}$-$\underline{\text{s}}$ / *teeth-ie-$\varnothing_{\text{PL}}$
f. sheep-$\varnothing_{\text{PL}}$ → ✓sheep-$\underline{\text{ie}}$-$\underline{\text{s}}$ / *sheep-ie-$\varnothing_{\text{PL}}$
g. fish-$\varnothing_{\text{PL}}$ → ✓fish-$\underline{\text{ie}}$-$\underline{\text{s}}$ / *fish-ie-$\varnothing_{\text{PL}}$

Forms like those in (7) but which retain umlaut are also possible, as (8) shows. Note that we still see the emergence of the default plural suffix in these cases:

(8) *Another possibility: Diminutive forces regular plural but umlaut persists*
a. mice-$\varnothing_{\text{PL}}$ → ✓mice-$\underline{\text{ie}}$-$\underline{\text{s}}$ / *mice-$\underline{\text{ie}}$-$\underline{\varnothing_{\text{PL}}}$
b. geese-$\varnothing_{\text{PL}}$ → ✓geese-$\underline{\text{ie}}$-$\underline{\text{s}}$ / *geese-$\underline{\text{ie}}$-$\underline{\varnothing_{\text{PL}}}$
c. feet-$\varnothing_{\text{PL}}$ → ✓feet-$\underline{\text{ie}}$-$\underline{\text{s}}$ / *feet-$\underline{\text{ie}}$-$\underline{\varnothing_{\text{PL}}}$
d. teeth-$\varnothing_{\text{PL}}$ → ✓teeth-$\underline{\text{ie}}$-$\underline{\text{s}}$ / *teeth-$\underline{\text{ie}}$-$\underline{\varnothing_{\text{PL}}}$

Diminutive /-i/ in general has an informal character, which excludes it from many forms of written English. This fact represents a challenge to systematic data collection. Nevertheless, the existence of such diminutive forms both with and without umlaut is revealed by speaker intuitions. They can also be found in informal writing, for instance, on the internet.[6]

This concludes the description of the data. In summary, the puzzle is as follows. The diminutive suffix forces use of the default plural suffix in irregular plural nouns that otherwise would not select it. Additionally, for nouns where plural umlaut would normally apply, the presence of the diminutive renders umlaut an option rather than a requirement. I argue that permitting late merger of the diminutive suffix explains these facts. In the next section, I provide background on late merge, and why we should expect it to be possible for the diminutive suffix. After this, we will be ready for the main analysis.

3 The possibility of late-merging the diminutive suffix

Research in syntax/semantics argues that certain constituents can be added to a structure at a relatively late point (Lebeaux 1991, Sauerland 1998, Nissenbaum 2000, Stepanov 2001, Fox 2002). Such late merger is often argued to be available only for syntactic adjuncts. This is because adjuncts are, unlike arguments and complements, not explicitly selected for. Rather they are "extra" modifiers that are not obligatory for a structure's interpretation, and can thus be introduced more flexibly. Many works arguing for late merger of adjuncts use binding principle C, and related effects, as diagnostics for merge timing. Principle C, for in-

[6] It is possible to find many examples of diminutive plurals both with and without umlaut through a Google search, using quotation marks to search for precisely matching instances of the search target. Here are some representative examples with working links at the time of writing (June 2026):

(i)
a. Love to eat them **mousies** (Link)
b. Look at the cute little sleeping **miceies**! (Link)
c. Silly **goosies** had fun "helping" with the mail today (Link)
d. Happy Tuesday from the **Geesies**! (Link)
e. Her **footies** were sticky and she stuck to my palm. (Link)
f. Frozen **Feeties** (Link)
g. I have sparkling **toothies** (Link)
h. The fish special! Nice **teethies**! (Link)

stance, rejects configurations where a referring expression such as a name is c-commanded by a co-referent antecedent (9):

(9) *Binding Principle C*
She$_1$ thinks that Mary$_{2/*1}$ is smart.

Evidence for late merge can be found by examining how principle C and syntactic movement interact. Consider what happens when a *wh*-moving phrase contains a referring expression. Normally, if the base position of the moved phrase is c-commanded by a subject that is co-indexed with the referring expression, the configuration is ungrammatical. This is because the referring expression in the *wh*-moved phrase violated principle C before movement occurred. This effect is one among several phenomena that fall under the general term *reconstruction*. This ungrammatical configuration in question is illustrated in (10a) below, where the moved phrase contains a referring expression in its complement PP. Importantly in contrast, the similar example (10b) is acceptable. The relevant difference between (10a) and (10b) is that in the latter, the referring expression is inside a relative clause of the moving phrase. Note that a relative clause is an adjunct:

(10) *Principle C in complement versus adjunct of wh-moved phrase*
  a. *Principle C violation for complement of moved phrase*
  *[Which picture [of John$_k$]]$_1$ does he$_k$ dislike t_1?
  b. *No principle C violation in relative clause adjunct of moved phrase*
  [Which cookies [that John$_k$ ate]]$_1$ did he$_k$ enjoy t_1?

In (10a), the PP in the complement of N is obligatorily merged in NP right away, before the containing *wh*-phrase moves. This creates a principle C violation at the very beginning of the derivation. In contrast, since relative clauses are adjuncts, the relative clause in (10b) can be merged after *wh*-movement over the subject. Consequently, the name in the relative clause was never c-commanded by the co-referential subject, and thus it does not incur a principle C violation:

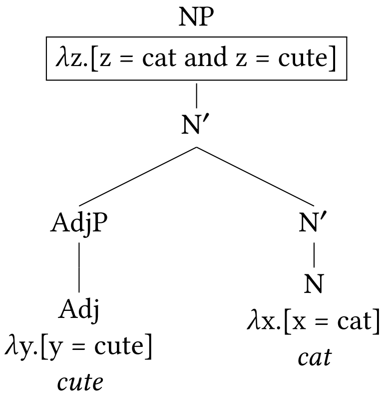
(11) *Simplified derivation for (10b)*
  a. *Step 1: Wh-movement occurs, relative clause not yet present*
  [Which cookies]$_1$ did he$_k$ enjoy t_1?
  b. *Step 2: Relative clause late-merged after wh-movement*
  [Which cookies [that John$_k$ ate]]$_1$ did he$_k$ enjoy t_1?

This is one representative piece of syntactic/semantic evidence for late merge of adjuncts. There is also precedent for late merger in morpho-syntax, from analyses of phenomena such as bracketing paradoxes (Pesetsky 1985, Nissenbaum 2000, Newell 2005, 2008). Combining these proposals, I argue for the possibility of late merger of the English diminutive /-i/.

3.1 Diminutive /-i/ as an adjunct/modifier that can merge late

Relative clauses and adjectives are widely considered adjuncts. Diminutive /-i/ can be regarded as having an essentially adjectival function. This strengthens the proposal that this diminutive morpheme is adjunct-like. Following Heim & Kratzer (1998), adjectives are predicates with the semantic type <e,t>, as are noun phrases. For instance, a noun *cat* with the semantic denotation λx.[x = cat], and an adjective *cute* with the denotation λy.[y = cute], are both predicates of type <e,t>. That is, they are predicates that take an entity, and determine whether it satisfies the truth-conditions involved in the concept in question. Follwing Heim & Kratzer, two elements with the same semantic type can be adjoined and semantically combined by the rule of Predicate Modification, as in (12):

(12) *Semantically uniting a noun and adjective*



The productive diminutive /-i/ has an adjective-like semantics in that it describes a noun as being small, cute, endearing, and so on. This could be characterized as a predicate λx.[x = small/endearing], which is type <e,t>, just like an adjective. This is the same semantic type that Heim & Kratzer also attribute to relative clauses. Relative clauses and adjectives are canonical instances of adjuncts, and the fact that the diminutive can be treated semantically the same reinforces the proposal that the diminutive suffix is adjunct-like. This result supports my proposal that diminutive /-i/ is capable of late merger.

There is also morpho-syntactic precedent for the hypothesis that diminutives are adjuncts/modifiers in some languages (Fábregas 2013, Gouskova & Bobaljik

2022). For instance, Gouskova & Bobaljik discuss noun gender as a distinguishing diagnostic. In German, diminutives formed with the suffix *-chen* always have the neuter gender, regardless of the gender that the base noun would have:

(13) *German diminutive overrides original gender*

Base	Gender	Diminutive	Gender
Wein ('wine')	Masc	Wein-**chen**	Neut
Feder ('feather')	Fem	Feder-**chen**	Neut

(Adapted from Gouskova & Bobaljik, ex. 3)

In contrast, in Italian, diminutive forms preserve the gender of the base noun:

(14) *Italian diminutive preserves original gender*

Base	Gender	Diminutive	Gender
ragazz-o ('boy')	Masc	ragazz-**in**-o	Masc
mamm-a ('mama')	Fem	mamm-**in**-a	Fem

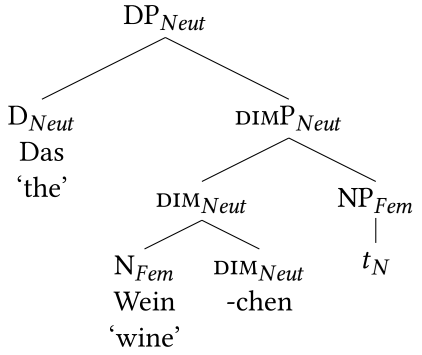
(Adapted from Gouskova & Bobaljik, ex. 4)

Drawing from previous work, Gouskova & Bobaljik (2022) argue that the German diminutive is not an adjunct, so its features syntactically project over, and thus suppress, those of the original noun. In contrast, the Italian diminutive is an adjunct, so it cannot overwrite the noun's original morpho-syntactic features.

I hypothesize that the English diminutive /-i/ is also an adjunct. To set the stage for this analysis, I will make explicit how we can diagram adjunct versus non-adjunct diminutives. I implement this analysis in Distributed Morphology, for which it is widely hypothesized that each morpheme in a complex word is a syntactic head, and that those heads form one word because they have been united by head-movement (Embick 2010, Arregi & Nevins 2012, Bobaljik 2012). In German, where the diminutive is not an adjunct, we should consider the diminutive *-chen* to project a phrase that dominates the NP, overwriting its gender, with N and the diminutive being united into one word via head-movement:[7]

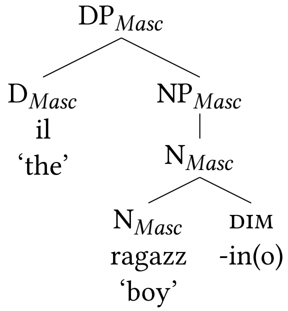
[7] In (15) below, N moves to the diminutive (dim) head, creating a complex head that must itself be labeled. A common assumption in syntactic theory is that the head which is moved to determines the label of the resulting head complex. For this reason, the constituent created by movement of N to the dim head is itself labeled dim. That second dim label then projects to form the higher diminutive phrase label (dimP). Such labeling in diagrams with head movement thus ensures an explicit representation. I employ the same labeling strategy for all diagrams in this paper that include head movement.

(15) *German diminutive projects a phrase* (DIMP = DiminutiveP)



In contrast, in Italian the diminutive does not project its morpho-syntactic features. Instead it adjoins to the noun, and it is the noun's features which take precedence. Thus the diminutive cannot overwrite the original gender:

(16) *Adjunct diminutive suffix in Italian*[8]



Such adjunction of an affix behaves similarly to adjunction of a phrase, such as an adjective or relative clause, to an NP. An NP to which adjunction occurs is still an NP, because the grammatical properties of adjuncts do not take precedence over the element that they adjoin to. In the Italian example above, the adjunct diminutive similarly does not project over the N that it attaches to, which thus retains the category label N. In this structure, note that the diminutive adjoins to N and creates a configuration which is the same as that typically formed by head-to-head movement. The only difference is that here the adjoining head is introduced by external merge rather than movement (=internal merge).

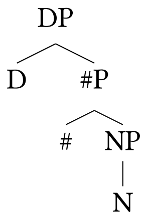
[8] A reviewer points out that in Italian, the masculine suffix *-o* is not a part of the diminutive as diagrammed here, since most analyses of gender marking in Romance take it to express a separate head in the functional structure of the noun. I acknowledge this fact, but maintain the simpler diagram in (16) for convenience.

Since I hypothesize that the English diminutive /-i/ is an adjunct, I will diagram it in the way shown above for Italian. However, to be prepared for the main analysis of English diminutive plurals, it is also necessary to consider the place of plurality in the morpho-syntactic representation. I describe this next, along with an initial illustration of late merge of the diminutive.

3.2 Implementing late merge of the diminutive in English plurals

For my analysis of English, I will assume the presence of a number phrase (#P) which sits between the NP and DP (Moskal 2015):

(17) *#P between NP and DP*

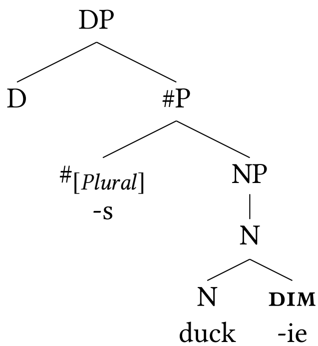


I will also stipulate that the N head always moves to the # head, uniting them into one word. Importantly, I will argue that the diminutive suffix /-i/ can adjoin to N at either an earlier or later point during this derivation.

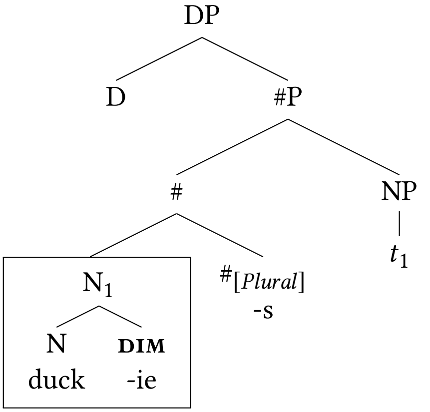
In the early-merge situation, I assume that the diminutive adjoins to N immediately, after which N and the diminutive it contains move to #, uniting them into one word. This is illustrated below with the regular plural noun *duck*:

(18) *Diminutive adjoins to N before head movement*

a. *Step 1: Diminutive adjoins to N*



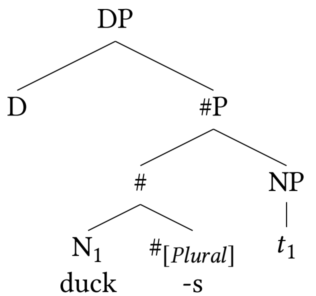
b. *Step 2: N containing diminutive moves to #*



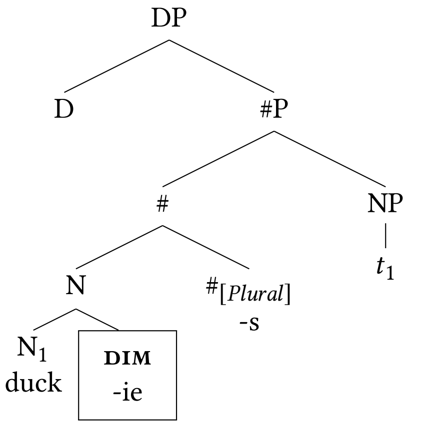
In contrast, in a derivation where the diminutive merges late, I assume that N moves to # first, after which the diminutive adjoins to N:

(19) *Diminutive adjoins late to N after head movement*

a. *Step 1: N moves*



b. *Step 2: Diminutive adjoins to N*



For the regular noun *duck*, merging the diminutive early or late makes no difference. However, for irregular plural nouns that normally show umlaut, the timing of merger matters, as the following sections argue. While nouns that normally show umlaut are most important here, along the way we will also straightforwardly account for irregular plural nouns for which umlaut is irrelevant.

4 Early merge derives diminutive plurals without umlaut

As we saw above, one possibility is for the diminutive to block all irregular plural morphology, resulting in use of the regular plural suffix, and a lack of umlaut in nouns that would otherwise have it:

(20) *Irregular plural morphology blocked by diminutive*

a. ox-en → ✓ox-ie-s / *ox-ie-en
b. mice-$\varnothing_{\text{PL}}$ → ✓mous-ie-s / *mice-ie-$\varnothing_{\text{PL}}$
c. geese-$\varnothing_{\text{PL}}$ → ✓goos-ie-s / *geese-ie-$\varnothing_{\text{PL}}$
d. feet-$\varnothing_{\text{PL}}$ → ✓foot-ie-s / *feet-ie-$\varnothing_{\text{PL}}$
e. teeth-$\varnothing_{\text{PL}}$ → ✓tooth-ie-s / *teeth-ie-$\varnothing_{\text{PL}}$
f. sheep-$\varnothing_{\text{PL}}$ → ✓sheep-ie-s / *sheep-ie-$\varnothing_{\text{PL}}$
g. fish-$\varnothing_{\text{PL}}$ → ✓fish-ie-s / *fish-ie-$\varnothing_{\text{PL}}$

To analyze this pattern, it is now necessary to be explicit about how Distributed Morphology functions. In this theory, syntactic structures are first built, only after which the underlying phonological forms of their terminal nodes are assigned. This assignment is determined by a list of (language-specific) Vocabulary Insertion (VI) rules, which define the correspondences between syntactic nodes/features and their underlying phonological forms. Some VI rules are context-neutral, and thus apply by default. These are termed *elsewhere* rules. Other VI rules are context-dependent. These apply under specific circumstances, and supersede default rules, yielding certain instances of allomorphy/suppletion.

In (21) below, I provide a set of VI rules relevant for the analysis of nouns that normally show umlaut in the plural. Since precise phonological detail is not necessary here, for convenience I use English spelling rather than phonemic representations. The purpose of the VI rules in (21) is to define the morphological properties of *goose/geese*. The rules (21a) and (21b) respectively encode that a singular version of # is expressed as -∅, while its plural version is realized by the form *-(e)s*. These are default rules about plurality that apply generally across English. In (21c) we have a default rule for the noun in question, which will realize it as *goose*. In (21d), there is a rule stating that this noun is realized as *geese* when adjacent to $\#_{[Plural]}$. Finally, in (21e) we have a rule stating that $\#_{[Plural]}$ is null when adjacent to that noun. In essence, these rules derive the form *geese* through bi-directional suppletion: A plural node adjacent to this noun requires the noun to take the form *geese* rather than *goose*, and simultaneously, this noun causes use of a null allomorph of $\#_{[Plural]}$ rather than *-(e)s*. Following Harley (2014), I assume that lexical roots are distinguished by abstract indices. The relevant noun in the rules in (21) below is thus marked with index 47. This ensures that only the noun typically realized as *goose* takes part in the alternations described in (21c-e).[9]

(21) *Some VI rules relevant for English plurals*

a.	$\#_{[Singular]} \leftrightarrow \varnothing$	(Default rule)
b.	$\#_{[Plural]} \leftrightarrow$ -(e)s	(Default rule)
c.	$N_{47} \leftrightarrow$ goose	(Default rule)
d.	$N_{47} \leftrightarrow$ geese / ___ $\#_{[Plural]}$	(Overrides 21c)
e.	$\#_{[Plural]} \leftrightarrow \varnothing$ / N_{47} ___	(Overrides 21b)

Rules of the same format can be defined for all nouns that show umlaut and exclude an overt plural suffix (*tooth/teeth, mice/mouse, lice/louse, foot/feet*, and so on). For instance, the following rules describe the behavior of *tooth/teeth*, which we might distinguish from *goose/geese* as involving a root N_{48}:

(22)

a.	$\#_{[Singular]} \leftrightarrow \varnothing$	(Default rule)
b.	$\#_{[Plural]} \leftrightarrow$ -(e)s	(Default rule)
c.	$N_{48} \leftrightarrow$ tooth	(Default rule)
d.	$N_{48} \leftrightarrow$ teeth / ___ $\#_{[Plural]}$	(Overrides 22c)
e.	$\#_{[Plural]} \leftrightarrow \varnothing$ / N_{48} ___	(Overrides 22b)

For a noun like *ox*, which does not show umlaut but instead uses a special overt plural suffix, a simpler set of rules is sufficient:

(23)

a.	$\#_{[Singular]} \leftrightarrow \varnothing$	(Default rule)
b.	$\#_{[Plural]} \leftrightarrow$ -(e)s	(Default rule)
c.	$N_{49} \leftrightarrow$ ox	(Default rule)
d.	$\#_{[Plural]} \leftrightarrow$ -en / N_{49} ___	(Overrides 23b)

A similar rule set captures the facts about nouns like *fish, sheep, shrimp*, and so on, which have no morpho-phonological distinction in the plural. The structure of these rules is the same as those for *ox*, but the suppletive plural triggered by the noun in this case is -∅:

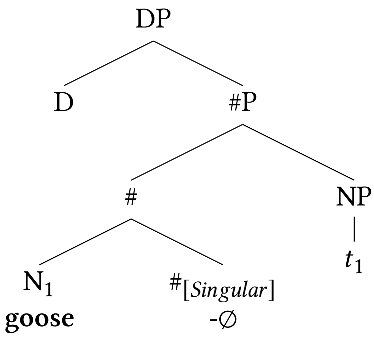
[9] In the tree diagrams that follow, I do not include root-distinguishing indices. I do, however, use numerical indices to mark a trace (*t*) and its matching moved element. These uses are distinct.

(24) a. $\#_{[Singular]} \leftrightarrow \varnothing$ (Default rule)
b. $\#_{[Plural]} \leftrightarrow$ -(e)s (Default rule)
c. $N_{50} \leftrightarrow$ fish (Default rule)
d. $\#_{[Plural]} \leftrightarrow \varnothing$ / N_{50} ___ (Overrides 24b)

In summary, the VI rules for nouns with umlaut like *goose/geese* are the most complex, involving bi-directional allomorphy. Similar rules are sufficient for nouns like *ox* and *fish*, though the rule sets for such nouns are simpler, since they do not show umlaut. Such VI rules provide the basis for this analysis.

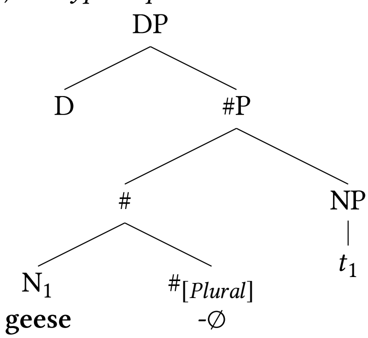
First, consider how the rules for *goose/geese* function when a diminutive is not present. In this situation, N will simply move to #. When # is singular its form is -∅, and it does not trigger umlaut, so the default form *goose* is assigned to N:

(25) *Plain singular "goose"*



If # is instead plural as in (26) below, it is again realized as -∅ due to the presence of this noun, as defined in (21e). At the same time, adjacency of that noun to $\#_{[Plural]}$ triggers the VI rule with umlaut, yielding *geese*, as defined in (21d):

(26) *Typical plural with umlauted "geese"*

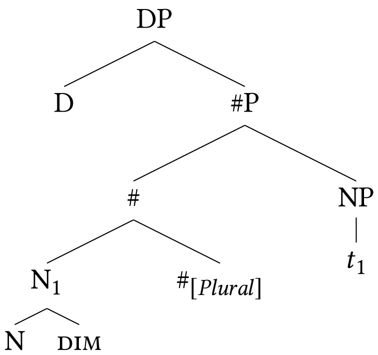


As described above, this is essentially a case of bi-directional suppletion: this N and $\#_{[Plural]}$ trigger use of context-specific VI rules for each other.

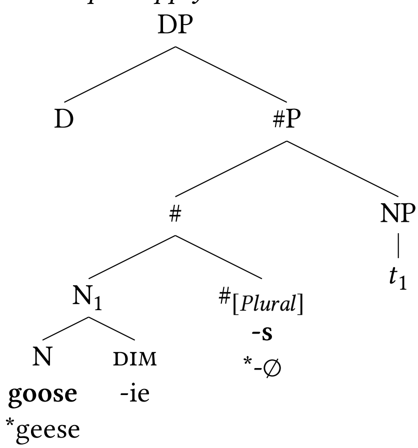
Importantly, much work has observed that the node that triggers a context-specific VI rule, and the node that undergoes it, must typically be adjacent (Embick 2010, Bobaljik 2012, Bobaljik & Harley 2017). Thus if another element intervenes between the N and $\#_{[Plural]}$, we expect suppletion to fail, causing their default forms to arise instead. Diminutive examples fitting this description can be found, as in (7) and (20) above. These are correctly generated by a derivation where the diminutive is merged early to N, followed by movement of N to #, after which VI rules apply. Since N and $\#_{[Plural]}$ are non-adjacent due to the diminutive intervening between them, they cannot receive their context specific forms, respectively *geese* and -∅. Rather they receive their default forms, *goose* and *-(e)s.* This is demonstrated in (27):

(27) *Intervening diminutive forces N and $\#_{[Plural]}$ to take default forms*

a. *Step 1: Merge diminutive, move N*



b. *Step 2: Apply VI rules*

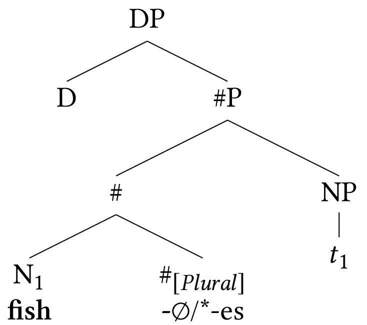


There is debate in the literature on whether context-sensitive VI rules that trigger suppletion/allomorphy require structural or linear adjacency between the nodes in question. For further discussion see Embick (2010), Bobaljik (2012), Lemon (2024). For the data in this chapter, hypothesizing a linear adjacency requirement straightforwardly captures the facts. This is particularly so if linear adjacency also registers the presence of null nodes (Nissenbaum 2000, Embick 2010), since this analysis importantly involves null plural nodes (see also section 6 below).

In contrast, a structural adjacency hypothesis has more potential problems. In (27b) above, for instance, where suppletion fails due to hypothesized lack of adjacency, $\#_{[\mathit{Plural}]}$ is arguably not structurally adjacent to the core N node where VI takes place. However, due to adjunction of the diminutive, there is a higher N label here which is sister to $\#_{[\mathit{Plural}]}$. Such multi-layered head complexes are common in representations of word formation. If we take this diagram seriously, the fact is that $\#_{[\mathit{Plural}]}$ is adjacent to a node bearing the features of the relevant N, which should be sufficient to trigger suppletion, at least for $\#_{[\mathit{Plural}]}$. Since this is not what occurs in reality, this cannot be the correct analysis. The linear adjacency theory does not have such an issue.

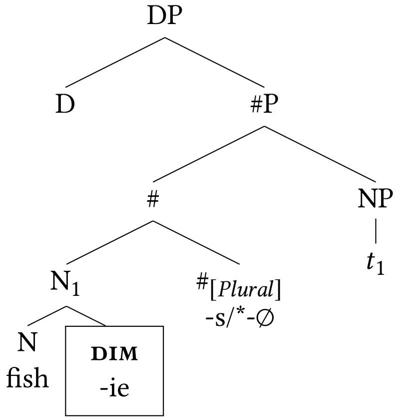
The form of analysis that we have seen in (27) can derive all the diminutive examples where irregular morphology is blocked, which we saw in (7/20) above, including *mousies*, *footies*, *toothies*, and so on. This analysis also straightforwardly applies to examples such as *oxies* and *fishies*, for which umlaut is irrelevant. The presence of the diminutive between the noun and the # node requires # to take on its default form rather than any context-sensitive one. For example, normally the N for *fish* triggers use of a null plural, as the VI rule in (24d) above describes:

(28) *An irregular plural without umlaut*



However, if the diminutive intervenes between the two, $\#_{[\mathit{Plural}]}$ must be realized by its default form instead, as we have seen in reality:

(29) *Irregular plural form blocked due to lack of adjacency with noun*



The same form of analysis works for a noun like *ox*, which normally triggers use of an overt suppletive plural suffix as in *ox-en*, except when the diminutive intervenes, yielding *ox-ie-s* rather than **ox-ie-en*.

Here we have considered early merge of the diminutive. Next, we will see that when it merges late instead, we correctly derive forms that contain the default plural suffix, while retaining umlaut if applicable.

5 Late merge derives diminutive plurals with umlaut

As (8) above showed, it is also possible for nouns that normally show plural umlaut to maintain it when the diminutive is used. Importantly, in this situation the default plural suffix still occurs, as (30) shows again:

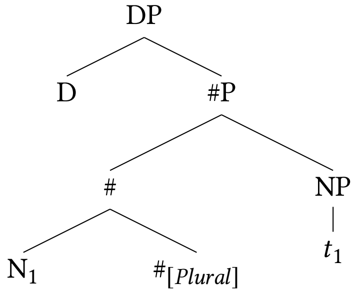
(30) *Another option: Diminutive forces regular plural but umlaut persists*

a. mice-∅$_{\text{PL}}$ → ✓mice-ie-s / *mice-ie-∅$_{\text{PL}}$
b. geese-∅$_{\text{PL}}$ → ✓geese-ie-s / *geese-ie-∅$_{\text{PL}}$
c. feet-∅$_{\text{PL}}$ → ✓feet-ie-s / *feet-ie-∅$_{\text{PL}}$
d. teeth-∅$_{\text{PL}}$ → ✓teeth-ie-s / *teeth-ie-∅$_{\text{PL}}$

This pattern presents an apparent paradox. Here N and $\#_{[Plural]}$ are not adjacent, and use of the null variant of $\#_{[Plural]}$ fails as expected. Nevertheless, the umlauted form of N is surprisingly still possible. We correctly predict this result by permitting late merge of the diminutive suffix immediately after the form of N is determined, as I will now demonstrate with *geese-ie-s*.

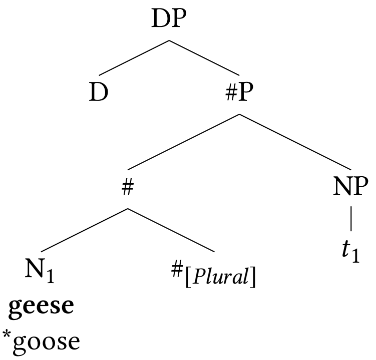
In this situation, first N moves to #, after which VI rules begin to apply:

(31) *Structure before VI*



Research in Distributed Morphology frequently argues that VI proceeds from the bottom to the top of a structure, starting at the root node (Embick 2010, Bobaljik 2000, 2012). This hypothesis will be essential here. Thus I assume that VI applies at N first. At this time, N is adjacent to $\#_{[Plural]}$, so VI must select the form with umlaut, which is *geese*:

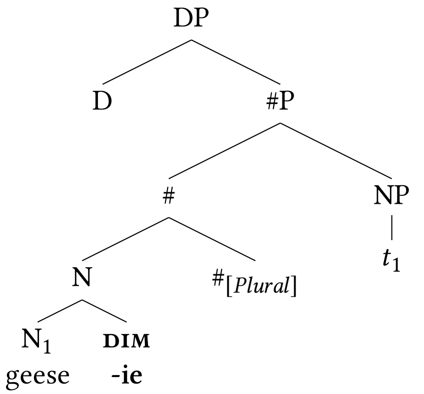
(32) *VI begins at N*



I argue that next, the diminutive late-merges to N. Assume that the diminutive node is subjected to its appropriate VI rule at this time.[10]

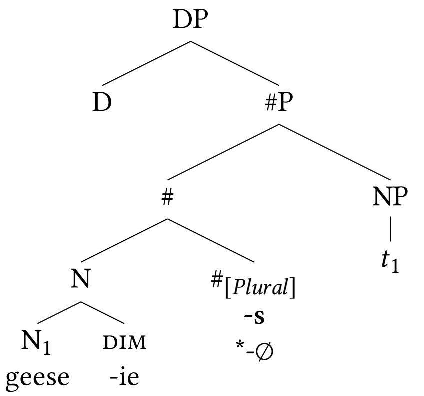
[10] Alternatively, we can assume that the diminutive was introduced in a separate workspace and spelled-out there, before being merged to the noun. See Piggott & Travis (2014) for an implementation in this vein for adjunction into words in Ojibwe.

(33) *The diminutive late-merges to N*



Next VI applies to $\#_{[Plural]}$. Since it is currently not adjacent to N due to the intervening diminutive, VI must select its default form *-(e)s*:

(34) *VI applies at* $\#_{[Plural]}$, *default form required*



We have thus derived an example where umlaut succeeds, though the default plural suffix is required. The same analysis can generate analogous examples we have seen above like *feeties, teethies, micies*, and so on.[11]

[11]This analysis has assumed that nouns like *mice, geese, teeth* and *feet* have a null plural suffix. This contrasts with the view in, for example, Bye & Svenonius (2012), who propose that irregular plurals like *mice* do not involve a null plural suffix. For Bye and Svenonius (p. 441), *mice* is a lexical entry linked to both the conceptual content of "mouse" and the plural node, which it simultaneously expresses as one portmanteau form. Such a theory is likely compatible with examples like *mousies*, given an additional plausible proposal that an intervening diminutive interrupts portmanteau formation, forcing the noun and plural node to be expressed separately by default forms. However, it is less clear how such a view can accommodate examples like *micies*, where we have the default plural suffix but persistence of umlaut.

The proposal that VI proceeds bottom-up, starting at N, is essential for this analysis. This is what makes it possible to apply VI at N and achieve umlaut, before establishing the form of $\#_{[Plural]}$. This analysis thus opposes theories permitting top-down VI, or everything-at-once VI. See Deal & Wolf (2017) and references therein for discussion about VI direction. Allowing top-down VI, for example, makes the incorrect prediction that it should be possible to have a silent $\#_{[Plural]}$ but no umlaut. This would generate examples that look exactly like diminutive singulars (*mousie, toothie, goosie*, etc.) but which are interpreted as plural. Such examples are unacceptable. The specific derivation is as follows:

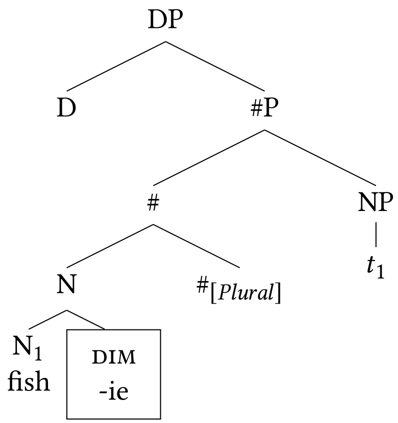
(35) *Undesirable derivation predicted by top-down VI*

1. Move N to $\#_{[Plural]}$ as usual.
2. Realize $\#_{[Plural]}$ as -∅ since it is adjacent to a relevant N.
3. Late-merge the diminutive between N and $\#_{[Plural]}$.
4. Assign N its default umlaut-less form, since at this time it is no longer adjacent to $\#_{[Plural]}$ due to the presence of the diminutive.

Maintaining that VI is strictly bottom-up avoids incorrectly predicting the possibility of this derivation for irregular plural nouns.

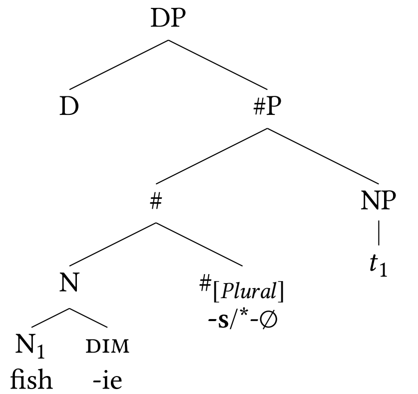
Allowing late-merge of the diminutive does not make incorrect predictions about irregular plural nouns that lack umlaut, such as *ox*, *fish*, and so on. Consider a situation where we have assigned the N for *fish* its appropriate form, and late-merged the diminutive right after:

(36) *The diminutive late-merges after VI at N*



After this late-merger, $\#_{[Plural]}$ is not adjacent to *fish*, so its default form *-(e)s* must be assigned, rather than -∅. This correctly derives *fishies*:

(37) *Default form of* $\#_{[Plural]}$ *required due to lack of adjacency*



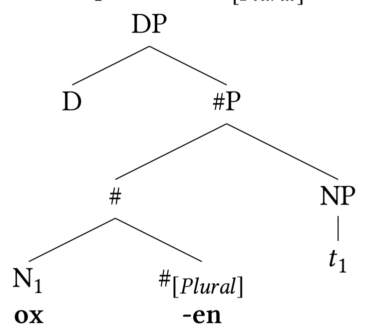
There is one more conceivable derivation to consider: late-merging the diminutive after both N and $\#_{[Plural]}$ have been realized. As I explain in the next section, this derivation over-generates, but we can exclude this possibility with an independently proposed constraint on late merge.

6 Preventing over-generation by even later merge

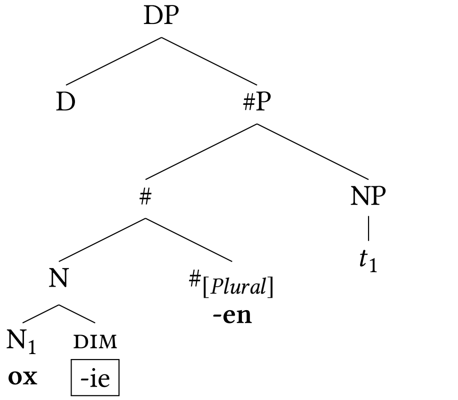
Nothing said so far rules out late-merging the diminutive between N and $\#_{[Plural]}$ after VI has applied to both of them. For instance, consider a derivation that expresses N and $\#_{[Plural]}$ as *ox* and *-en*, afterwards late-merging the diminutive between them. This would generate an ungrammatical example **ox-ie-en* where a context-sensitive plural allomorph is used, despite not being adjacent to the triggering node:

(38) *Undesirable later late-merge*

a. *Step 1: N and* $\#_{[Plural]}$ *undergo VI while adjacent*



b. *Step 2: Diminutive merges between*

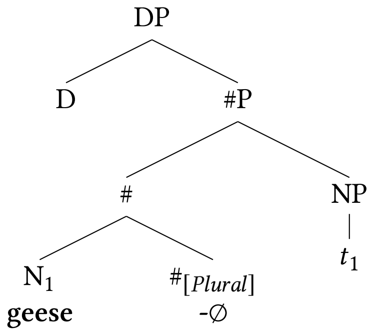


The same process could derive an ungrammatical form like **fishie* (as a plural), where the plural node is realized as -∅ while adjacent to *fish*, followed by late merger between them.

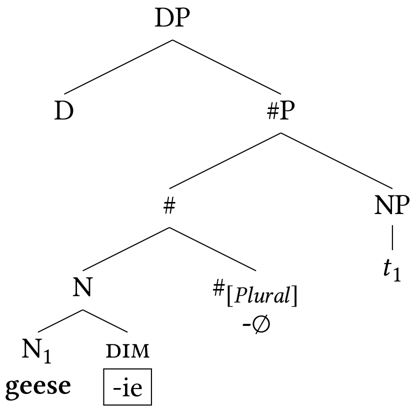
Similarly for a noun like *goose/geese*, we wrongly predict the possibility of realizing N and $\#_{[Plural]}$ using the special VI rules for each (*geese*, -∅) while they are adjacent, before merging the diminutive between. This would incorrectly derive *geesie* as a plural noun:

(39) *Another undesirable derivation*

a. *Step 1: N and $\#_{[Plural]}$ undergo VI while adjacent*



b. *Step 2: Diminutive merges between*



In summary, a derivation with late-merge that is even later over-generates, by incorrectly predicting diminutive plurals such as **ox-ie-en* and **geesie*. Therefore we must rule out this type of derivation.

For independent reasons that are rooted in syntactic phenomena such as extraposition, Nissenbaum (2000) argues that late merge is constrained by what he terms the *linear edge condition*. This states that it is possible to late-merge at the linear edge of structure that has been morpho-phonologically realized, but it is not permitted to late-merge between nodes that have been spelled-out. Nissenbaum (pg. 203-206) argues that this constraint can be observed in phrase-level syntax, and also in morphology to some extent.

With this constraint in mind, notice that in the representative undesirable derivations in (38) and (39) above, late-merge of the diminutive between N and $\#_{[Plural]}$ after they have been subject to VI violates the linear edge condition. This is because in these cases, late-merge applies structurally below the spelled-out $\#_{[Plural]}$, and thus between two nodes that are already morphologically realized. Importantly, the fact that $\#_{[Plural]}$ is realized as -∅ in (39) does not change this. Nissenbaum explicitly argues that the linear edge condition also applies for structure that has been realized with null morphology.[12] Thus it is not possible to late-merge between spelled-out nodes, regardless of whether VI assigned them overt or silent morphology. In general, the linear edge condition clearly parallels the *phase impenetrability condition* (Chomsky 2000, 2001), which posits that spelled-out material is not available for manipulation by syntactic operations. In

[12]This hypothesis sits well with independent proposals in morpho-syntax that un-pronounced nodes are visible at Phonological Form, and thus subject to linearization (Arregi & Nevins 2012, Haugen & Siddiqi 2016).

summary, if late-merger cannot penetrate structure that has undergone morphophonological expression, we correctly rule out the undesirable derivations described above.

7 Extension: Late merge and long distance allomorphy

I have adopted the hypothesis that contextual allomorphy/suppletion can only be triggered when the node undergoing allomorphy, and the element that serves as the contextual trigger, are linearly adjacent. While this is a frequently-observed state of affairs, there are instances of apparent allomorphy between non-adjacent nodes. Predicting the cross-linguistic limits on allomorphy and related processes is a topic of debate and ongoing research. For an overview see Moskal & Smith (2016) and Kastner & Moskal (2018). While some version of an adjacency condition on allomorphy seems likely, the analysis of this chapter in fact reveals one way in which long-distance allomorphy might nevertheless emerge.

The examples in this chapter where we find nouns that display umlaut, despite not being adjacent to the plural element that is presumably the trigger for it, involve something analogous to long-distance allomorphy. I argued that this is derived by the noun taking on an allomorph with umlaut while adjacent to the triggering plural node, after which the diminutive is late-merged between the two, with the plural node undergoing VI afterward. Importantly, derivations of this general format can derive long-distance allomorphy in at least certain cases.

A representative example of long-distance allomorphy comes from Basque. As Bobaljik (2012) p. 156-8 discusses, Basque shows long-distance suppletion for the adjectives "much" and "good". For instance, the adjective "much" is *asko* in the plain form (40a). However, it becomes *gehi* when followed by the comparative suffix *-ago* (40b). Interestingly, this suppletion occurs even when the morpheme *-xe* ("a little") intervenes (40c):

(40) *Long-distance suppletion in Basque*

a. asko
"much"

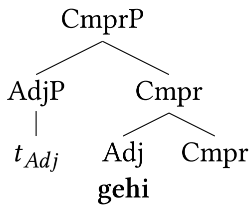
b. gehi-ago
much-CMPR
"more"

c. gehi-**xe**-ago
much-a.little-CMPR
"a little more"

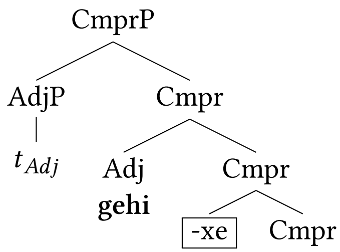
Late-merge of *-xe* can derive this pattern. Following Bobaljik (2012), comparative adjectives involve a comparative phrase (CmprP) that dominates the adjective. Assuming that word formation requires head-movement, in a Basque example like (40c) above, first the adjective head will move to the comparative head. This is illustrated in (41a) below, in which VI at this adjective has occurred. Being adjacent to the comparative head at that time VI occurs, this adjective "much" receives its suppletive form *gehi* rather than its default *asko*. If next VI proceeds to express the comparative head with its form *-ago*, we would derive example (40b) above. However, I propose that before VI spells-out the head of CmprP, late merge of the morpheme *-xe* can occur. Bobaljik proposes that *-xe* semantically modifies the comparative and therefore attaches to it, essentially as in (41b) below. After this late merger, the derivation ends with VI at the head of CmprP, realizing it as *-ago*, thus deriving *gehi-xe-ago* as in (41c):

(41) *Long-distance allomorphy by late merge in Basque*

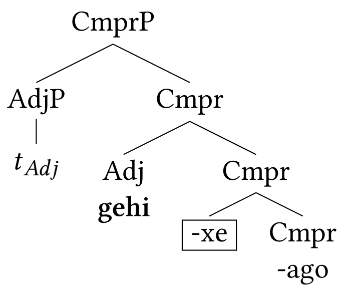
a.



b.



c.



Since *-xe* is a degree modifier, it is adjunct-like, as Bobaljik argues. Thus *-xe* is a good candidate for a morpheme capable of late-merge.

In section 4, I argued that a linear adjacency requirement on allomorphy is best suited to the analysis of the English diminutive data. The above analysis of Basque is also compatible with such a constraint, since it achieves allomorphy before the needed linear adjacency is interrupted by late merge. However, Bobaljik (2012) argues for what is essentially a structural adjacency account for Basque. In short, the analysis is that the adjective undergoes allomorphy due to being the sister of the Cmpr node that dominates *-xe* and the actual Cmpr terminal where VI occurs. This leaves us with multiple potential analyses for Basque, which it is beyond the scope of this paper to adjudicate between.[13] Since the cross-linguistic nature of adjacency in allomorphy is a broad and complex topic, I will not speculate further on this here.

The point of this section is that the appearance of long-distance allomorphy could, in certain cases, feasibly be derived by late merger of the intervener. Such an analysis cannot account for all cases of long-distance allomorphy, since not all intervening elements in such scenarios will be semantic modifiers/adjuncts. A more likely, though weaker hypothesis, is that only some cases of long-distance allomorphy are epiphenomena of late merger. I leave this topic to future research.

8 Concluding remarks

The diminutive suffix /-i/ shows complex behavior when combined with irregular plural nouns. This diminutive always forces use of the default plural suffix, but optionally allows plural umlaut to persist. I have argued that these facts emerge from (a) the proposal that diminutive morphemes in English are adjuncts which can merge late, (b) a linear adjacency requirement on context-sensitive allomorphy, (c) the hypothesis that VI proceeds bottom-up from the root, and (d) a ban on late-merging linearly within spelled-out material. In this context, the optionality of umlaut in plural diminutives is reduced to the optionality of late-merge. This analysis involves the application of a well-supported proposal in syntax, namely

[13] In the context of a linear constraint on allomorphy, late merge of *-xe* can indeed generate *gehi-xe-ago* as described above. However, if it were also possible to early-merge *-xe*, we may predict the availability of a form *asko-xe-ago* with no suppletion. Analogously to what I have proposed for English diminutives, we expect allomorphy to be blocked when an intervener merges early, and for seemingly long-distance allomorphy to be derived when the intervener merges late. As far as I know *asko-xe-ago* is not possible in Basque. Bobaljik's structural analysis does not face this complication. This issue would dissolve if late merge is in fact obligatory (Stepanov 2001), but this proposal causes problems for the analysis of English, where optionality of late merge is essential. It would also be consistent with my arguments to hypothesize that the English diminutive is either an adjunct or not, and that when it is an adjunct, it must late-merge.

late merger, in the domain of morpho-phonology. We indeed expect to find such effects if the foundation of morphology is syntactic structure, as argued in theories such as Distributed Morphology. More generally, this analysis demonstrates how phenomena that are peripheral or seemingly idiosyncratic can turn out to be theoretically informative, and are waiting to be discovered even in extremely well-documented languages such as English.

8.1 A remaining puzzle about *kitty* and *bunny*

There are two noun suppletion phenomena in English triggered by the diminutive itself. First, the diminutive causes an irregular vowel change of *cat* to *kit*, yielding *kitty*. Second, the diminutive appears to cause suppletion of *rabbit* to *bun*, yielding *bunny*. The potential regular diminutive forms that we might have expected are unavailable:

(42) *Potential noun suppletion triggered by the diminutive*

   a. cat ([kæt]) → kitty ([ˈkɪ.ti]) / *catty
   b. rabbit ([ˈɹæ.bɪt]) → bunny ([ˈbʌ.ni]) / *rabbitie, *rabbie

If the diminutive suffix can be late-merged after N has undergone VI, as the main analysis of this chapter proposed, we expect suppletion to *kitty* and *bunny* to be optional, contrary to fact. For instance, if the diminutive were merged to the N for *cat* before VI begins, adjacency of that N to the diminutive might cause its suppletion to *kit*, as in *kitty*, matching what we see in reality. In contrast, if the diminutive were late-merged after N undergoes VI, N would be realized as *cat*, yielding the unacceptable **catty*. The same process would yield either *bunny* or **rab(bit)ie* from *rabbit*, though only *bunny* is actually permitted. It is therefore evident that the main analysis of this chapter over-generates in these cases. I note that *catty* has an adjectival use which may interfere with its ability to serve as a diminutive. Additionally, **rabbitie* would violate the requirement to be disyllabic, but truncating it to **rabbie* does not resolve the problem (see footnote 4 on truncation in diminutives).

It is not entirely obvious that *kitty* and *bunny* are morphologically decomposable. If these are in fact mono-morphemic, they could involve expression of both N and the diminutive node simultaneously (Bye & Svenonius 2012, Svenonius 2016, Haugen & Siddiqi 2016), though it is unclear how this would interact with potential late-merger. Another option may be to simply regard *kitty* and *bunny* as morpho-syntactically unrelated to *cat* and *rabbit* respectively, despite their semantic connection. Ultimately, I must leave this puzzle for future research.

Abbreviations

DIM = diminutive	NEUT = neutral
FEM = feminine	PL = plural
MASC = masculine	SG = singular

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